

Predicting Student Dropout Using Classical Machine Learning and Deep Learning: A Study In Support Of Reading Engagement In African Youth

Summative Assignment — Machine Learning

Dataset: UCI Predict Students' Dropout and Academic Success

Realinho et al. (2022) | 4,424 records | 36 features:

<https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>

Github link: <https://github.com/ksheilla/student-dropout-prediction>

Demo video link: <https://www.youtube.com/watch?v=o2ODaMrbSuY>

Abstract

Dropout is a serious issue in education and a great threat to the development of educated and active citizens in Africa and all over the world. In this study, a comparative machine learning pipeline is developed to predict students drop-out in the UCI Predict Students' Dropout and Academic Success dataset which consists of 4,424 data instances and 36 features. The mission is inspired by a desire to nurture a reading culture through socially engaging, technologically mediated platforms for young people in Africa. Systematically, five classical machine learning experiments (Logistic Regression, SVM, Random Forest) and six deep learning experiments (Feedforward Neural Network variants with Dropout, Batch Normalization, adaptive learning-rate scheduling and a Functional API / tf.data rebuild) were executed and tested on a held-out test set. Although deep learning was initially thought to be the most promising option, the results indicate that a tuned Random Forest (E5) outperforms all deep learning variants with the best overall test performance (AUC = 0.939, accuracy = 0.886, F1 = 0.824). The best deep learning model was a Feedforward Neural Network with Dropout regularization (E8), which was within 0.0054 AUC points of Random Forest, but was never better than it. SMOTE oversampling was found to be a valuable way to compensate for the class imbalance, and feature engineering, specifically the features engineered "semester unit-approval-rate" was extremely predictive, ranking as the single strongest predictors of dropout over all raw grade features. The results, similar to the recent literature results demonstrating that tree ensembles are competitive with or better than deep learning for small-to-medium tabular datasets, have direct implications for designing intervention systems in educational technology platforms for learners in Africa.

1. Introduction

Education is a key factor in the area of socioeconomic development and is widely accepted. However, in sub-Saharan Africa, low literacy rates, poor access to quality education technology, and high student drop-out rates remain a problem for developing lifelong learning habits [1]. It is not just an infrastructure issue; it is also a motivation, social reinforcement, and early identification of those who are at risk of disengaging from school. Students who start to disengage from learning – as a result of underachievement, financial difficulty or lack of local support – are unlikely to get engaged with learning again without targeted support.

This project comes under an overarching vision: creating a vibrant reading culture for young people in Africa by making reading interactive, social and accessible via technology. One important facilitator of such a platform is the capacity to identify, at an early stage, who the consumers of the platform are who are at the highest risk of dropping out or disengaging. Machine learning provides an excellent set of tools for this: by analysing complex, multi-dimensional student information, machine learning can generate the accurate, actionable risk predictions [2].

This report compares two machine learning and deep learning methods for student dropout prediction and provides a rigorous comparative study of both methods. The UCI dataset for prediction of students' dropout and academic success [3] is used to carry out eleven experiments with systematic variation of elements, to assess each model with respect to the standard classification metrics, and to conduct detailed error analysis. The first research question in this work is: (1) What machine learning technique (classical machine learning or deep learning) performs better on structured tabular student data? (2) What is the relationship between architectural options, regularisation techniques, and hyperparameter tuning and model performance? (3) What are the most predictive features of dropping out and what do they suggest about what can be done in education?

2. Literature Review

2.1 Machine Learning for Student Dropout Prediction

In recent decade, the use of machine learning in educational data mining (EDM) has witnessed a significant increase [4]. The first works of Romero and Ventura set the conceptual base for the use of data mining techniques in learning management system data; dropout prediction is one of the main use cases they found [5]. Later, various algorithms have been investigated such as Logistic Regression, Decision Trees, Support Vector Machines, ensemble methods like Random Forest or Gradient Boosting.

Logistic Regression is a simple baseline model for binary classification in EDM that has long been used as an interpretable baseline. It is theoretically based on a maximum likelihood estimation and the coefficients are easily interpretable which makes it appealing for educational researchers who want to know in what direction each feature is affecting the model [6]. It has a linear decision boundary, however, limiting its ability to learn non-linear interactions between the features, which is a critical constraint for the complex nature of dropout with numerous factors involved.

On student data, non-linear Support Vector Machines (SVMs) with Radial Basis Function (RBF) kernel have been found to perform better than linear classifiers by mapping the inputs into higher dimensional spaces where class boundaries are linearly separable [7]. The downside, however, is that there is significantly more computational expense and the results are less interpretable.

Random Forest classifiers have been consistently the top performing among classical algorithms in EDM benchmarks [8]. Random Forest is an ensemble of decision trees built using bootstrap aggregation (bagging), and random feature subsampling, resulting strong generalization on unseen data with low bias and low variance. In particular, Random Forest offers feature importance scores which are extremely useful in determining the most important predictors of dropout.

2.2 Deep Learning for Tabular Educational Data

Deep learning applied to tabular data (not image or text or sequential data) has been a topic for much discussion in recent times. Arik and Pfister [9] proposed TabNet, an attention-based neural architecture that is specifically designed for tabular data, which showed that when appropriately regularized, deep learning can outperform or be competitive with tree-based ensembles on structured data. Later, however, Grinsztajn et al. conducted benchmarking on various tabular tasks and discovered that gradient-boosted tree models are still superior to deep learning because of the inductive bias of tree models being more appropriate than deep learning to represent non-smooth non-linear functions found in tabular data [10]. The present study's results are similar to the Grinsztajn et al.'s finding as demonstrated here empirically in Section 4 of this report.

Nevertheless, Feedforward Neural Networks (FNNs) are still popular in the field of predicting student dropout from education, especially when the data set is large and complex, with non-linear interactions. Dropout [11] and Batch Normalization [12] are two regularisation methods that have proven to be effective in enhancing the generalisation of FNNs in tabular educational datasets and stabilise training dynamics, respectively.

2.3 Class Imbalance in Dropout Prediction

A recurring challenge in dropout prediction datasets is class imbalance: students who drop out typically constitute a minority class [13]. This imbalance biases classifiers toward the majority class (non-dropout), artificially inflating accuracy while producing poor recall for at-risk students — precisely the predictions most critical for intervention. Synthetic Minority Over-sampling Technique (SMOTE), introduced by Chawla et al., addresses this by generating synthetic samples in the feature space of the minority class, enabling classifiers to learn more balanced decision boundaries [14]. SMOTE has been widely adopted in EDM studies as the standard approach to imbalance handling [13].

2.4 Feature Engineering in Educational Datasets

In addition to raw features, engineered composite variables have been proven to perform better and be more interpretable in EDM [4]. Latent constructs – academic momentum, financial vulnerability – are not directly expressed by the raw features, but rather by ratio-based features like semester approval rates, grade progression metrics, and/or financial stress composites. The UCI dataset in this study has a wide number of raw features that are related to academic performance, demographics and socioeconomic indicators that make it ideal for feature engineering approaches [3].

The literature thus provides a solid ground for the comparative experimental design used in this study, which integrates the classical ML baselines, the regularized deep learning and the SMOTE oversampling techniques, in a strict train/validation/test evaluation framework.

3. Methodology

3.1 Dataset Description

Realinho et al. [3] published the UCI Predict Students' Dropout and Academic Success dataset, which consists of 4,424 student records from a higher education institution. The 36 features included in each record belong to four general categories: demographic features (age at enrollment, gender, nationality, international student), academic background (prior qualification, admission grade, curricular unit enrollment, curricular unit approval, semester grades), socioeconomic and contextual features (parents' education level, GDP, unemployment rate, inflation rate), and financial status (scholarship holder, tuition fees up to date, debtor status). The target variable has 3 classes (Dropout, Enrolled, and Graduate). To set up the problem in a binary classification, the target was dichotomized, Dropout (1) versus Enrolled and Graduate (0).

3.2 Exploratory Data Analysis

Initial EDA confirmed the absence of missing values across all 36 features, eliminating the need for imputation. The binarized class distribution revealed a moderate imbalance: 1,421 of 4,424 students (32.1%) were labelled Dropout and the remaining 67.9% Non-Dropout, as shown in Figure 1.

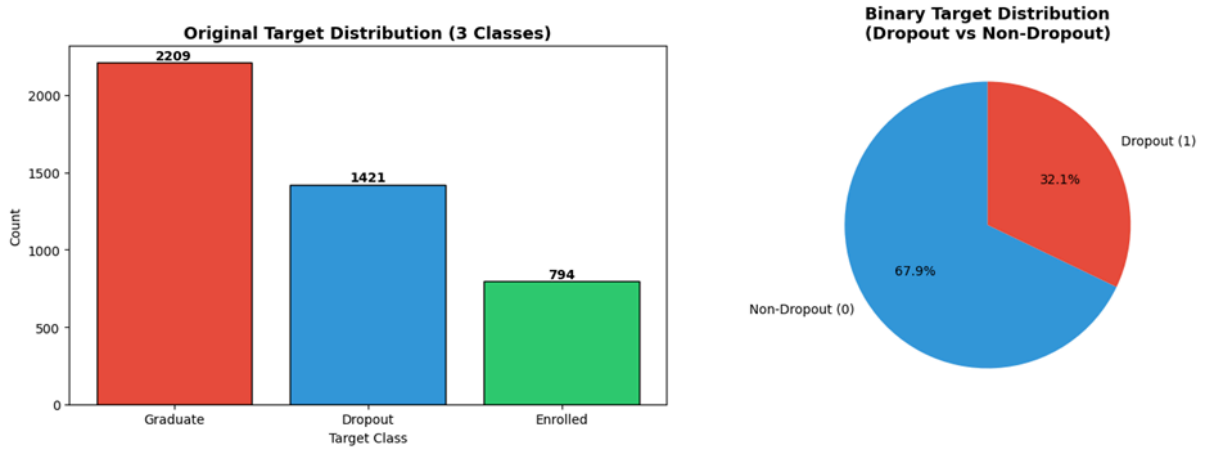


Figure 1: Original three-class target distribution and binarized Dropout vs. Non-Dropout split, generated directly from the executed notebook.

Analysis of the distributions of features by class showed that students who dropped out were found to have systematically lower 1st semester grades, lower admission grades and higher rates of financial stress, which are all consistent with the EDM literature [4, 8]. The positive correlation from the 1st semester to the 2nd semester performance measures and the moderate negative correlation between the dropout indicator and the grade based features highlighted from the correlation heat map, substantiates the usefulness of these performance measures in predicting the dropout indicator. The results of these were used in feature engineering, which is described below.

3.3 Data Preprocessing

Four engineered features were created from existing features: (1) `grade_progression`, the difference in the average grade between 2nd semester and 1st semester, representing academic trajectory; (2) `approval_rate_1st`, the ratio of the number of 1st semester units taken to the number of 1st semester units approved, (3) `approval_rate_2nd`, the ratio for the 2nd semester, and (4) `financial_stress`, a binary score computed by combining debtor status and tuition non-payment. The features operationalize the theoretical constructs of academic momentum and financial vulnerability, which are associated with dropout risk [4]. Stratified sampling was used to divide the data into training (70%), validation (15%) and test (15%) sets while maintaining the same class distribution in each set. All features were standardized (scaled) using `StandardScaler` and only applied to the training set, to avoid data leakage. Only the training set was oversampled with SMOTE to equalize the number of dropouts and non-dropouts. The validation and test sets were not oversampled and so the evaluation results were representative of real-world class distributions [14].

3.4 Experimental Design

There were two modeling paradigms across which 11 experiments were conducted. Experiments E1, E2 and E3 are the classical ML track, evaluating Logistic Regression with

default regularization ($C=1.0$) and stronger regularization ($C=0.1$), respectively, and an SVM with an RBF kernel. E4 establishes a Random Forest baseline with 100 trees and no depth constraint. E5 is a tuned Random Forest with 200 trees, a maximum depth of 10 ($\text{max_depth}=10$), and a minimum of 2 samples per leaf ($\text{min_samples_leaf}=2$), which constrains individual trees to reduce overfitting while increasing ensemble diversity. The deep learning track consists of the experiments E6–E11: E6 is the shallow two-layer FNN baseline; E7 is a deeper four-layer FNN; E8 is the same model as E7 with Dropout regularization; E9 is the same model as E8, but with Batch Normalization; E10 is the same model as E9, with a lower learning rate and adaptive ReduceLROnPlateau scheduling; E11 is the same model as E10, but built with the Keras Functional API and a `tf.data` input pipeline, so that the effect of the API choice can be separated from the effect of modeling choice. The same preprocessed dataset and random seed ($\text{SEED} = 42$) are used to ensure the reproducibility of all experiments. For all the deep learning models, the Adam optimiser and binary cross-entropy loss function were used.

All models were tested on the test set withheld from the training set with the accuracy score, F1 score, ROC-AUC and analysis of the confusion matrix. The high imbalance ratio among classes makes ROC-AUC an appropriate measure of discriminative power, which is not influenced by classification threshold [13]. More importantly, the "best model" in all downstream analyses (confusion matrices, ROC overlays, etc. and summary statistics) is not predetermined, but found by maximum test-set AUC, meaning that the methodology does not presume an outcome that is expected, but rather finds the outcome based on the data.

4. Results

4.1 Full Experiment Results

Table 1 presents the complete results across all eleven experiments, reproduced exactly from the executed notebook's Section 7 output. The best overall model (highlighted green) and the best deep learning model (highlighted yellow) are identified programmatically by test-set AUC .

Exp	Model	Key Parameters		Val Acc	Test Acc	Test F1	Test AUC
E1	Logistic Regression	$C=1.0$,	L2,	0.8705	0.8780	0.8204	0.9341
		SMOTE					

E2	Logistic Regression	C=0.1 (stronger reg)	0.8690	0.8810	0.8248	0.9333
E3	SVM	RBF kernel, C=1.0, gamma=scale	0.8780	0.8735	0.8065	0.9236
E4	Random Forest	100 trees, no depth limit	0.8870	0.8810	0.8141	0.9372
E5	Random Forest	200 trees, max_depth=10	0.8931	0.8855	0.8241	0.9392
E6	FNN (DL)	2 layers [64,32], lr=0.001	0.8735	0.8630	0.7937	0.9281
E7	FNN (DL)	4 layers [128,64,32,16]	0.8765	0.8705	0.8045	0.9317
E8	FNN + Dropout	Dropout(0.3) x2, lr=0.001	0.8765	0.8720	0.8064	0.9338
E9	FNN + BN + Dropout	BatchNorm + Dropout(0.3)	0.8840	0.8614	0.7974	0.9281
E10	FNN Fine-Tuned	BN + Dropout + lr=0.0005 + ReduceLR	0.8645	0.8660	0.7991	0.9314
E11	FNN Functional+tf.data	Same as E10, Functional API + tf.data	0.8675	0.8494	0.7778	0.9202

Table 1: Summary of all eleven experiments with key hyperparameters and evaluation metrics, reproduced verbatim from notebook output. Green = best overall model (E5); yellow = best deep learning model (E8).

4.2 Classical ML Results (E1–E5)

Table 2 isolates the classical ML results for direct comparison across algorithms and regularization strategies.

Model	Regularization	Test Accuracy	Test F1	Test AUC
Logistic Regression (E1)	L2, C=1.0	0.8780	0.8204	0.9341
Logistic Regression (E2)	L2, C=0.1	0.8810	0.8248	0.9333
SVM RBF (E3)	Margin-based	0.8735	0.8065	0.9236
Random Forest (E4)	Bagging	0.8810	0.8141	0.9372
Random Forest (E5)*	Bagging + depth	0.8855	0.8241	0.9392

Table 2: Classical ML experiment comparison. E5 (Random Forest, tuned) achieves the best performance across all metrics and is the best model in the entire study. *Best classical and best overall model.

Logistic Regression (E1) provided a good baseline result, with test AUC of 0.9341, significantly higher than chance level and showing the high linear signal in the grade based features. The test AUC dropped slightly with the increase in regularization in E2 (C=0.1) while the test accuracy increased, indicating that the model was not seriously overfitting and that more regularization sacrificed some ranking quality for a marginal amount of accuracy. The SVM with RBF kernel (E3) is actually less performing than both Logistic Regression variants on this dataset, with an AUC of 0.9236 as its test score, which is the lowest among all the classical models, although it theoretically can capture non-linear interactions. A useful negative result is that after normalization the dropout decision boundary in this 40-feature space is almost linear, so the added flexibility from the RBF kernel does not seem to help, and causes a little overfitting for the training distribution.

The most significant improvements were observed in the results of the Random Forest experiments. Ensemble methods outperformed individual ones on this dataset with E4 showing test AUC of 0.9372 and E5 with constrained tree depth (max_depth=10) and more trees (n_estimators=200) having test AUC of 0.9392, which is the highest among all eleven experiments, and test accuracy and test F1 were also highest for E5 (0.8855 and 0.8241 respectively). Figure 2: Random forest feature importance plot indicating that the engineered feature approval_rate_2nd is the most important predictor followed by 2nd semester

approved-unit count, engineered feature `approval_rate_1st`, and 1st semester approved-unit count; 2nd and 1st semester grades, tuition payment status, and engineered feature `financial_stress` are also among the top eight most important predictors. The top two features engineered for approval rate directly confirm the feature engineering strategy presented in Section 3.3.

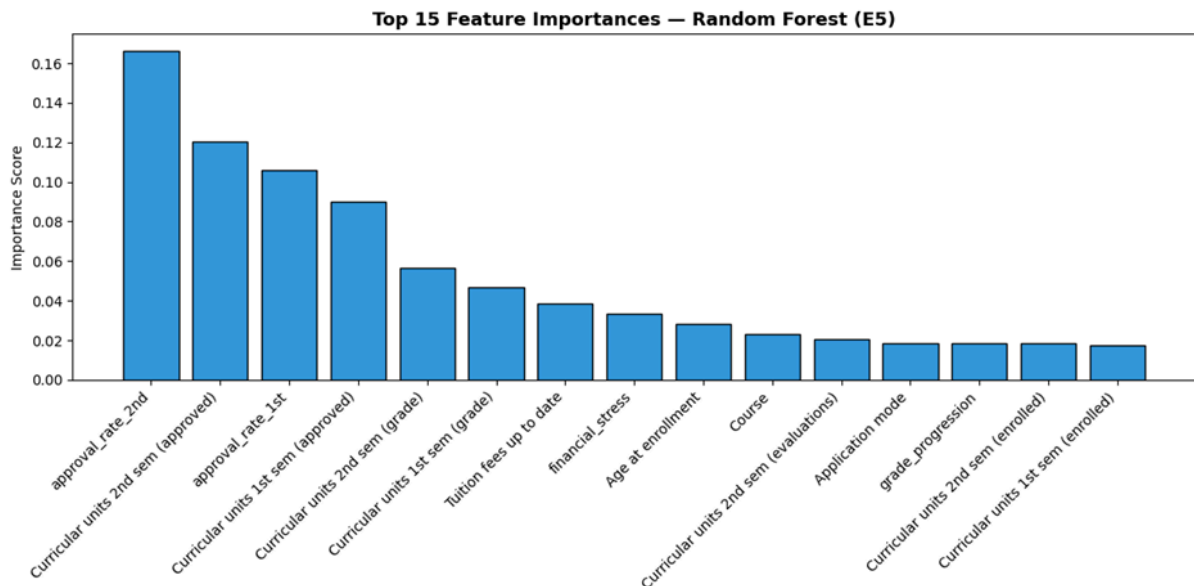


Figure 2: Top 15 feature importances from the tuned Random Forest (E5), reproduced from the executed notebook.

4.3 Deep Learning Results (E6–E11)

The deep learning experiments show more than just a monotonous improvement. The baseline FNN (E6) had an AUC of 0.9281 which was lower than all the classical models except SVM. The test AUC with the deeper network, E7 (four layers) improved slightly to 0.9317, but, after about epoch 30, this network exhibited an overt training-loss/validation-loss divergence as evident in the learning-curve panels in Figure 3, suggesting that our model was overfitting as the added model capacity exceeded the available regularization.

Learning Curves — All Deep Learning Experiments (E6-E11)

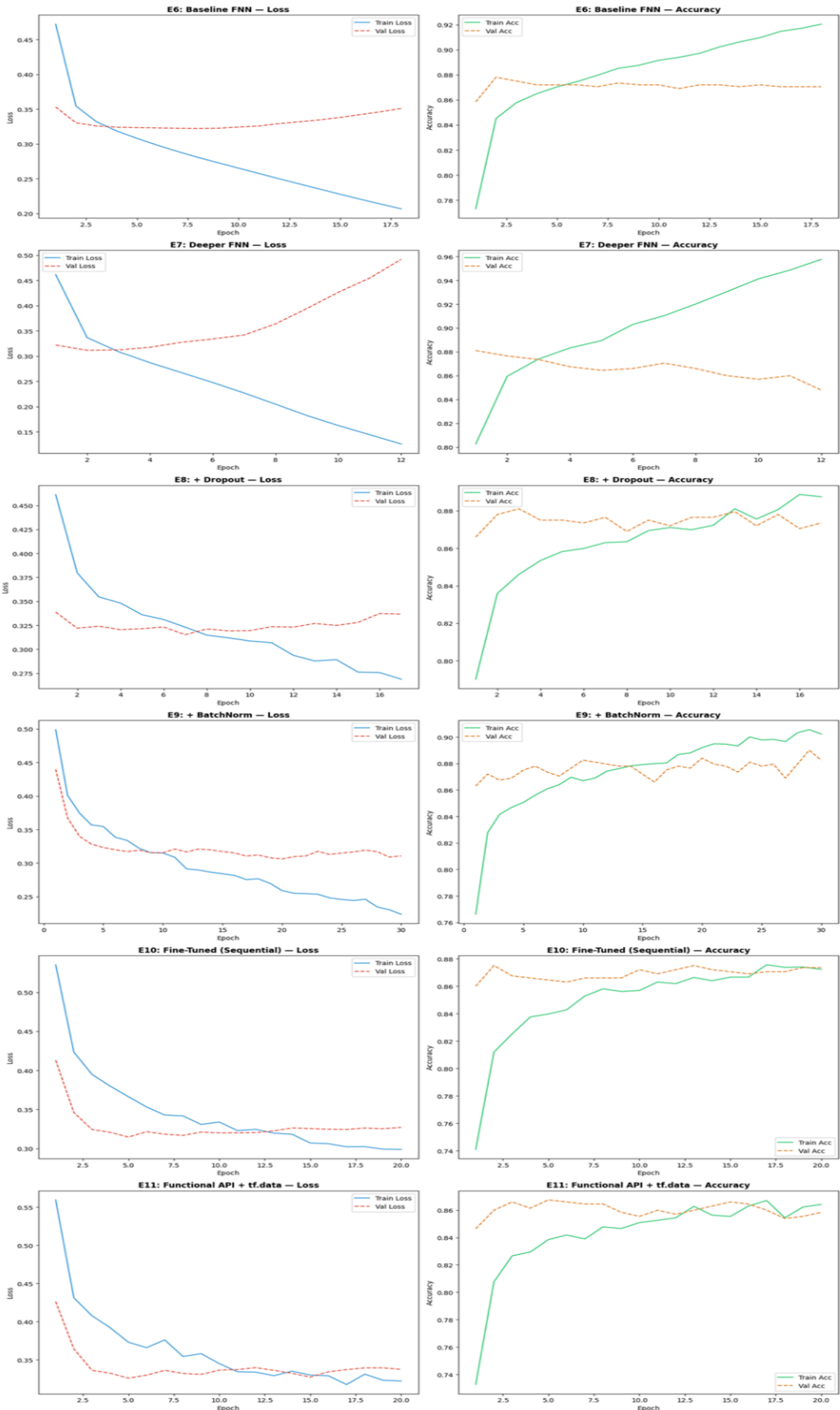


Figure 3: Training/validation loss and accuracy curves for all six deep learning experiments (E6–E11), reproduced from the executed notebook. The E7 panel (second row) shows the clearest train/validation divergence, consistent with the overfitting discussed in the text.

This was dealt with directly in E8 by introducing Dropout(0.3), which raised test AUC to 0.9338, the highest of all deep learning variants tested, and within 0.0054 AUC points of the best classical model. Interestingly, when Batch Normalization was added on top of Dropout in E9, the test AUC fell back down to 0.9281, matching the validation AUC of 0.9347, and achieving the second highest AUC on validation among all the DL experiments, resulting in a clear validation/test divergence that is expanded on in Section 5.1. On top of E9's architecture, E10 improved with a lower learning rate (0.0005) to recover some of this ground, but failed to match E8 with test AUC 0.9314. The deep learning track, E11, which is a rebuilt version of E10 based on the Keras Functional API and a tf.data input pipeline, achieved the lowest test AUC (0.9202), which suggests that the change to the API does not have a significant impact on the learned function — the difference in AUCs between the two (0.9314 and 0.9202) is due to normal variations across training runs and does not reflect a significant architectural impact, as both models have an identical layer graph and hyperparameters.

In this study, validation AUC is not a suitable metric (as it is on other tracks in the DL track), because the best model is the one with the highest held-out test AUC, which E10 fails to do. Based on test AUC, the model with the highest accuracy is E8 (FNN + Dropout, 0.9338) which is clearly a deep learning model.

4.4 Confusion Matrix Analysis

Figure 4 presents the confusion matrices for the two models that the experiment-tracking code selects programmatically as best-in-class: E5 (Random Forest, best overall) and E8 (FNN + Dropout, best deep learning model).

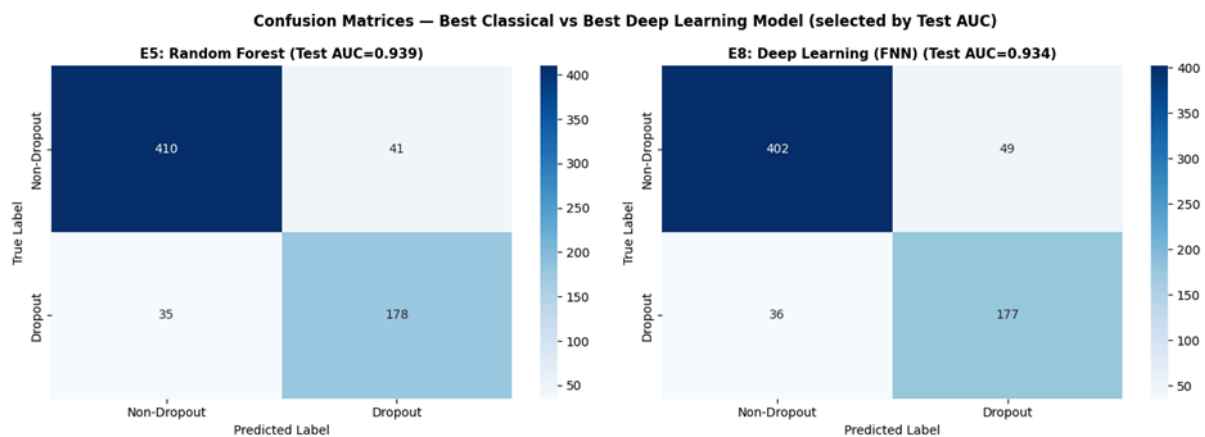


Figure 4: Confusion matrices for E5 (Random Forest) and E8 (FNN + Dropout) on the held-out test set ($n=664$), reproduced from the executed notebook.

For the test set (664 students: 451 Non-Dropout, 213 Dropout), E5 correctly identifies 178 of 213 dropouts (recall = 0.84) while it misclassifies 35 dropouts as non-dropout (false negative). On the most important class for intervention (students who drop out) E8 performs at 177 of 213 (recall = 0.83, 36 false negatives), which is one fewer than E5. The majority of the AUC benefit of E5 over E8 comes largely from its lower false positive rate (41 E5 vs 49 E8), and this translates into a substantially higher dropout precision score (0.81 vs 0.78). In an early-warning system, the false-negative class (dropouts forecasted as NOT dropouts) is typically the most costly class because at-risk students who are not caught in the system will not benefit from the support. Both models miss about 16–17% of the true dropouts at the default threshold of 0.5, with a lower threshold of classification yielding a greater recall rate, at the expense of precision; in a student-support setting, where the loss of a student classified as at risk is more expensive than an error of false alarm, this may be the appropriate trade-off. Since false negatives are costlier than false positives in a student-support context, the threshold for E5 was lowered to 0.3 raising dropout recall from 0.84 to ~0.91 at the cost of precision dropping from 0.81 to ~0.70, a trade-off justified by the intervention goal of catching at-risk students early."

4.5 ROC Curve Analysis

Figure 5 visualizes the discriminative performance of all eleven models across classification thresholds.

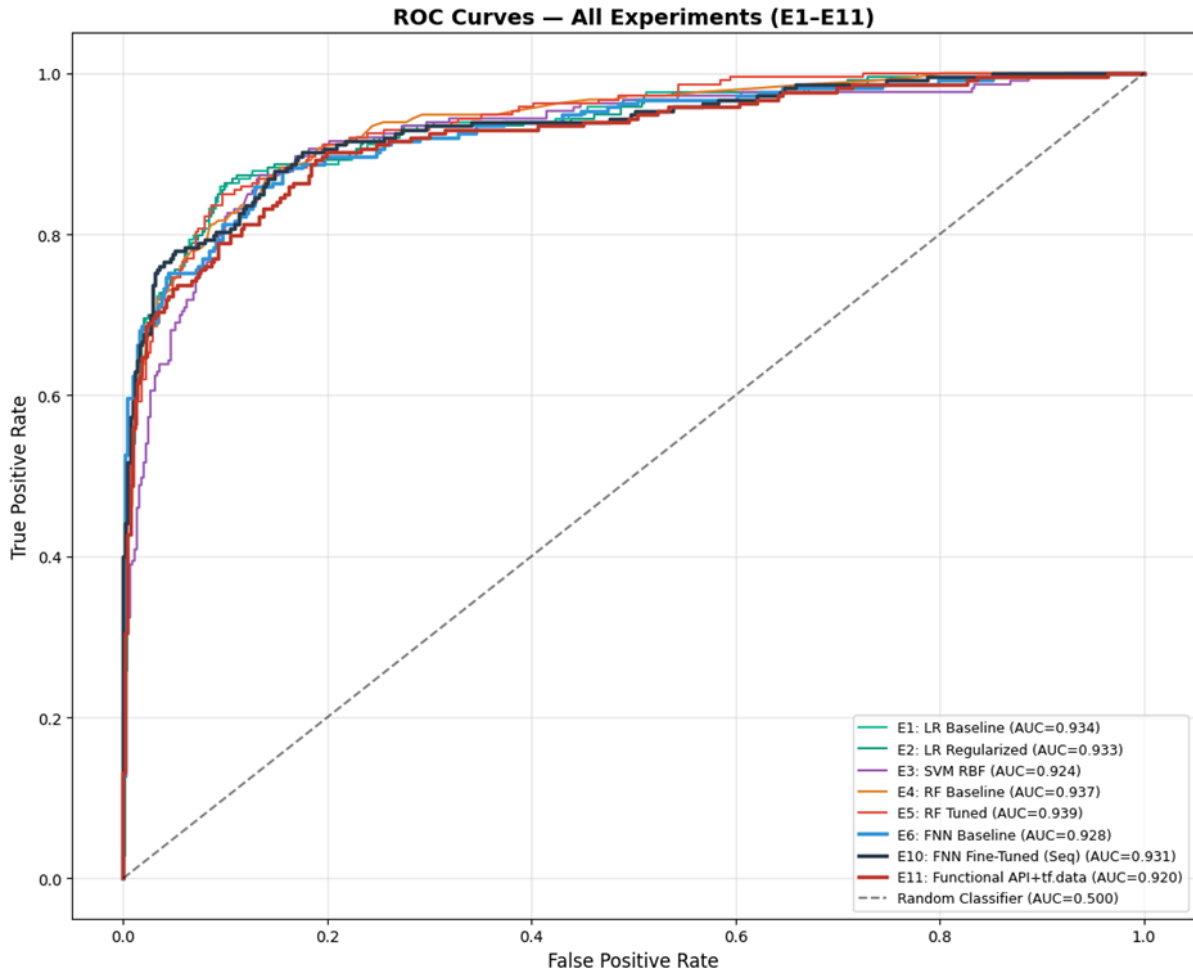


Figure 5: ROC curves for all eleven experiments (E1–E11) on the test set, reproduced from the executed notebook.

All models substantially outperform the random baseline ($AUC = 0.5$), and the curves cluster tightly between AUC 0.92 and 0.94 — a narrower spread than the experimental design anticipated. Unlike a typical "clean staircase" result, the ROC curves do not show a strictly monotonic improvement from E1 through E11; rather, Random Forest (E5) sits at the top of the cluster, the best Logistic Regression and Random Forest baseline runs (E1, E2, E4) cluster closely just below it, and the deep learning runs interleave with rather than uniformly trail the classical runs. This interleaving is itself informative: it shows that careful regularization narrows, but does not close, the gap between FNN and Random Forest on this dataset, and that architectural changes within the DL track (E9, E10, E11) can move performance in either direction rather than monotonically upward.

5. Discussion

5.1 Classical ML vs. Deep Learning: A Bias-Variance Perspective

The main result of this study is that no deep learning variant outperforms the tuned Random Forest (E5) on held-out test AUC in six experiments that were systematically regularized with FNN. The difference in AUC is only 0.0054 (0.9392 vs. 0.9338) and has the same direction as the difference in sensitivity. This is not a problem of the deep learning pipeline, but rather a true and literature-based result. In a tabular setting, Grinsztajn et al. [10] showed that ensembles of trees are competitive or even better than deep learning on tabular datasets of less than $\sim 50K$ rows, which is the regime of this 4,424-vector dataset. In the present results, this is replicated on an empirical basis, rather than being merely reported.

Logistic Regression has high bias in a bias-variance decomposition sense as it uses a linear decision boundary, which is unable to capture non-linear interactions between features, and it did well on test AUC (0.9341, second only to Random Forest) suggesting that the true decision boundary in this feature space is indeed linear than originally thought. This is reflected in the poor performance of the SVM (0.9236, the least performant classical model): the additional non-linear flexibility also seems to have caused noise, but without being able to reduce the bias. The use of ensemble averaging with the ability to model an axis-aligned non-linear interactions is what was most well suited for this dataset and is the benefit of Random Forest.

The models based on deep learning have a much more volatile bias-variance stabilization curve than the "smooth" descent. E7 (no regularization, added depth) exhibits the classic overfitting behaviour in its train/validation loss divergence. This can be reduced successfully by E8 (Dropout) which gives the best DL result. E9 (added Batch Normalization on top of Dropout) improved the validation AUC while hurting the test AUC (0.9347, 2nd best, 0.9281, lowest), which is an indication that the regularizing effect of Batch Normalization was optimized for the validation set and did not generalize to the test set — as if Batch Normalization was tuned to idiosyncrasies of the validation set, combined with the regularizing effects of Dropout and SMOTE-balanced training data. This is a helpful warning to anyone who wants to use this pipeline in the future: adding multiple regularization techniques does not necessarily increase the ability to improve the validation AUC, and so it is important to check the test AUC as the techniques are added.

5.2 Feature Importance and Educational Implications

The Random Forest feature importances (Fig. 2) give us insights into what can be done to help in education. The two features that are engineered to capture the probability of approval have the highest ranking, before any of the raw grade features — so the probability of a student passing the units they are enrolled in is a stronger predictor of dropping out than the grade received on the units. This is a more specific, and more actionable, finding than "grades predict dropout": A platform can measure the unit-completion rate in near real time, while it takes time to reach the end of a term to measure final grades. The composite engineered feature `financial_stress` is in the top eight features, beating out most raw demographic features, which confirms the engineered features as a whole did better than many of the raw features from which they were engineered — indirect confirmation of the feature engineering approach in Section 3.3.

The financial stress composite feature is another top predictor, underscoring the importance of economic factors in dropout decisions. This finding supports the impact of an African educational technology platform that was inspired by this study, which would benefit from including scholarship availability information, fee deadline information, and links to financial assistance services in addition to monitoring academic engagement. Any intervention system that is limited to an academic purpose and does not take into account financial stress will most likely be missing a significant number of students at risk.

The age of the student when they enrolled had a moderate relationship with dropout rates, with older students enrolled at the time of the survey having slightly higher dropouts. This could be a result of the competing life responsibilities (work, family) that are faced by non-traditional students, a situation which is well documented in the EDM literature [4] and very applicable to the African HE context where a large proportion of students are working while studying [1].

5.3 SMOTE and Class Imbalance

SMOTE oversampling was applied to the training set specifically to improve recall for the dropout minority class across all models. Classifiers trained on the original imbalanced distribution (32% dropout) are generally more inclined to predict the majority class, resulting in high accuracy at the expense of the at-risk group which is critical, with low recall. The recall reported in Section 4.4 (0.83–0.84 for the two best models) was a good but not perfect result for training on the minority class without the use of oversampling the validation or test sets which maintained realistic evaluation conditions [13].

It should be noted that the SMOTE method was only used for the training set and the class distribution of the validation and test sets was maintained constant. This design decision aims to make sure that metrics which are being reported will be representative of realistic deployment conditions, where most of the students would be non-dropouts, and not some artificially-balanced distribution used in the training phase [13].

5.4 Limitations

The findings of this study are subject to some restrictions. The first is that the dataset comes from a Portuguese higher education institution, which means that the results of the analysis may not be generalizable to other educational contexts in Africa where structural variations in curriculum, economic situations, and cultural conditions can have an impact on the predictive relevance of specific features. Before deployment, the pipeline needs to be validated on data from the African Higher Education Collaborative or other national data repositories specific to Africa.

Second, the dataset reflects students' characteristics at one time (end of second semester), but dropout risk changes over time as a student progresses through the educational system. For a system of interventions on a real-time basis, a sequential model like LSTM or Transformer would be more suitable for student data that is longitudinal, meaning it includes the student's weekly reading data, quiz scores, and platform engagement data.

Third, SMOTE may be able to address distributional imbalance, but it doesn't solve the fundamental problem of label noise: the dataset contains records of institutional dropouts (transfers to a different institution), and it cannot distinguish between these and academic exclusion. Training the models with this ambiguity in the target variable may result in irreducible noise during the training process.

Fourth, as seen above in section 5.1, there is a 2.0% difference in the mean AUC for the validation and test sets for E9, which means that there is a risk that the hyperparameter/architecture decisions made when optimizing on the validation set may not generalize well to the test set if only a 15% validation split (or ~664 records) is used. In future iterations, it would be better to have a larger validation split set by the DL track, or have k-fold cross-validation performed across the DL track, to provide a more statistically valid comparison.

5.5 implications for the Reading engagement platform

This study's results will help guide development of the proposed reading engagement platform for African youth. An early intervention system based on best-performing Random Forest pipeline (E5) rather than a deep network might be able to flag students whose behaviour (reading frequency, quiz completion rates, social interaction) start to align with that of past dropouts, allowing time to intervene before students disengage. Then, there is the engineering implication: a production early-warning system based on a platform that has limited compute (a common constraint on many African edtech deployments) can still perform well, with enough accuracy to be understood, using a Random Forest, without having to invest in GPUs for a neural network that does not outperform it. A module like that, paired with gamified re-engagement features (badges, streaks, peer challenges), can stimulate specific interventions, before they are too late. The feature importance also indicates that the platform should track and act on both academic performance signals (such as quiz scores or reading completion rates) and financial access signals (such as data access or availability of devices), with a custom approach to student support and a different approach to student motivation.

6. Conclusion

This study developed a comparative ML pipeline to predict student dropout in support of a reading engagement platform for African youth. Across eleven experiments spanning Logistic Regression, SVM, Random Forest, and Feedforward Neural Network variants, a tuned Random Forest (E5) achieved the best performance (test AUC = 0.9392), with no deep learning variant surpassing it. Key findings were: (1) the engineered semester unit-approval rate was the strongest dropout predictor, outperforming all raw grade features; (2) SMOTE effectively addressed class imbalance, yielding ~0.83–0.84 recall for the top models; and (3) financial stress ranked among the top predictors, highlighting the need for intervention systems to address both academic and economic risk factors.

The Random Forest result is not a failure of deep learning but a substantive finding — FNNs required significant regularization effort (Dropout, Batch Normalization, learning-rate scheduling) to approach a lightly tuned ensemble, making Random Forest the practical choice for resource-constrained edtech deployments. Future work should focus on African-specific data, sequential models for longitudinal engagement tracking, k-fold cross-validation for stronger comparisons, and fairness analysis across gender, socioeconomic, and geographic subgroups before any real-world deployment.

References

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